

WORLDLINK COMMUNICATIONS PVT. LTD.

ENVIRONMENT DAY 2026 | CORPORATE SOCIAL RESPONSIBILITY INITIATIVE

Landscaping Proposal

Manjushree Park & Chobhar Community Forest

Three-Site Plantation Plan, Environment Day 2026



Satellite context of the project area at Chobhar Gorge, Kirtipur. The polygonal outline (upper) traces the community forest cleared zone proposed for Site 2; the rectangular outline (lower, along Dakchinkali Road) marks the Manjushree Park strip housing the proposed Site 1 Children's Park. Site 3 follows the road frontage between the two outlines.

Prepared for Worldlink Communications Pvt. Ltd.
In coordination with Manjushree Community Forest User Group (CFUG)
Location Chobhar Gorge, Kirtipur Municipality, Bagmati Province, Nepal
Coordinates 27.66° N, 85.29° E (approx. 1,350 m a.s.l.)
Approval Confirmed by the CFUG President, Manjushree CFUG
Initiative Environment Day Plantation 2026

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1 Background

This document presents a comprehensive landscaping proposal for a plot of land within **Manjushree Park**, Chobhar, Kirtipur Municipality, Kathmandu Valley. The proposal has been prepared in conjunction with **Worldlink Communications'** Environment Day plantation initiative, aimed at:

- Enhancing green cover and ecological sustainability.
- Beautifying the public recreational space through strategic fruit-tree and ornamental selection.
- Creating an interactive, edible landscape for park visitors and local wildlife.

Manjushree Park holds significant cultural and ecological importance. Established to restore the landscape scarred by a now-defunct cement factory at Chobhar Gorge, the park today serves as a peaceful recreational retreat for both residents and tourists. It is named after *Manjushree*, the Buddhist bodhisattva credited in legend with draining the ancient Nagdaha lake to create the Kathmandu Valley by cutting the Chobhar Gorge.

2 Study Area Description

2.1 Location & General Setting

Location	Chobhar Gorge, Kirtipur Municipality, Bagmati Province, Nepal.
Road Frontage	Dakchhinkali Road, a major highway running NW to SE through Chobhar.
Coordinates	Approximately 27.66° N, 85.29° E.
Altitude	Approximately 1,350 m above sea level (mid-hill zone).

2.2 Key Physical Features

Observations drawn from satellite imagery and field reconnaissance, mapped to the three sites in Figure 1:

- **Dakchhinkali Road** runs NW–SE through the site, separating Site 2 (north of the road, inside the community forest) from Site 1 (south of the road, inside Manjushree Park). It is a two-lane highway planned for widening to a **23 m corridor**, which directly drives the Site 3 design.
- **Site 1** (Children's Park): a roughly rectangular cleared plot within Manjushree Park, measuring approximately **64.6 m (E–W) × 10.6–12.6 m (N–S)**, with a flagstone-paved entrance / gate at the east end.
- **Site 2** (Community Forest): a cleared kidney-shaped zone within the surrounding forest, bordered on all sides by mature canopy that will act as a permanent windbreak and ecological buffer.
- **Site 3** (Roadside): the narrow linear strip immediately along Dakchhinkali Road, sitting within the future 23 m right-of-way. Overhead power and utility lines follow this corridor, constraining mature height.
- Gently sloping terrain (approx. 5–8% gradient) descends from the road toward the forested gorge below; no terracing is required.
- A roundabout and park entrance feature is visible at the north-western end of the site, and the dense



Figure 1. Proposed plantation sites at Chobhar, overlaid on satellite imagery (legend at right of figure). **Site 1** (yellow) is the small rectangular plot south of Dakchhinkali Road, marked at “Manjushree Park”, earmarked for the Children’s Park. **Site 2** (orange, kidney-shaped) is the cleared zone north of the road within the community forest, designated for the edge fruit/flowering plantation. **Site 3** (olive/dark green, long curving strip) follows the Dakchhinkali Road frontage and is the candidate for low shrub planting only, owing to the planned 23 m road widening.

gorge forest provides a natural backdrop.

- The scenic backdrop of Chandragiri and Champadevi hills frames the site to the south-west.

2.3 Climate & Soil Context

Kathmandu Valley falls within a **humid subtropical to warm temperate** climate zone. The area receives approximately **1,400 mm** of rainfall annually, concentrated between June and September (monsoon). Winters are mild with occasional frost at this altitude.

Soils in the Chobhar area are typically **sandy loam to clay loam**, moderate in organic matter, and slightly acidic to neutral (pH 5.5–7.0), well suited for a wide range of fruit and ornamental trees. Soil may have been historically affected by the cement bunker; visual inspection suggests these effects are not currently significant, but a basic pH and texture test prior to planting is recommended.

3 Methodology & Plant Selection

3.1 Site Assessment Approach

The following engineering and horticultural criteria were applied to evaluate species suitability:

- Altitude suitability (mid-hill: 1,000–2,000 m).
- Root system type (deep vs. shallow, critical near structures and paved areas).
- Canopy spread vs. available spacing.

- Risk classification (road hazard, utility-line interference).
- Water requirement and drought tolerance.
- Ecological benefit: soil binding, pollinator support, bird habitat.
- Community benefit: edible fruit, shade, aesthetics.

3.2 Revised Three-Site Zoning Framework

Based on field reassessment, the original three-zone framework (A, B, C) within a single strip was found to be **infeasible**: the strip is too narrow to support distinct front, mid, and rear belts. The proposal has therefore been restructured into a **three-site plan**, each with a distinct function and design priority (see Figure 1).

Site	Location	Function
Site 1	Cleared zone within Manjushree Park, earmarked for a Children's Park (~64.6 m × 11 m)	Natural Playground centred on log play, sand pits and curvilinear pathways, framed by slender evergreens on the north and a flowering/fruit sensory edge on the south.
Site 2	Adjacent Community Forest, cleared zone north of Dakchhinkali Road	Edge plantation of fruit/flowering trees with low hedges for site demarcation and ecological buffer.
Site 3	Roadside strip directly along Dakchhinkali Road (future 23 m road expansion corridor)	Low shrub plantation only, or no plantation, to preserve sight-lines and accommodate the future widened road.

Table 1. Three-site framework. All three sites have been confirmed by the CFUG President for Environment Day planting.

Design priority. Where any earlier zoning concept conflicts with the detailed Children's Park (Site 1) design described in Section 4, the Children's Park design takes precedence. Sites 2 and 3 are supporting plantations around that core.

4 Site 1: Children's Park — Natural Playground

4.1 Design Intent

Site 1 is the project's centrepiece: a cleared rectangular zone within Manjushree Park (approximately **64.6 m long × 10.6–12.6 m wide**) developed as a **Natural Playground for children**. The design deliberately avoids synthetic play structures and plastic toys in favour of timber, sand, stone and living plants. All internal geometry is curvilinear; rigid straight lines are excluded.



Figure 2. Detailed plan for the Site 1 Children’s Park. The plot measures **64.60 m (E–W) × 10.59–12.61 m (N–S)**; all dimensions on the figure are in metres. The main **Gate** sits at the east end. The **northern edge** is lined with slender conical evergreens (“slender trees for securing visibility”) interleaved with low spherical shrubs. The central **Natural Playground** (annotated with inset photographs of log balancing trails, sand pits and timber climbing frames) occupies curvilinear turf islands separated by organic-shaped sand/gravel zones and a paved walkway. The **southern edge** is a sensory band of flowering and fruit trees (guava, peach, plus pink and white seasonal bloomers) with a low manicured hedge line at ground level. Rustic log **benches** are positioned along the walkway and near the gate.

4.2 The Play Zone: Natural Playground

Positioned centrally within the park, the play area uses natural materials to stimulate physical agility, balance and imaginative play without relying on synthetic structures.

Log Balancing Trails Stepping stumps and fallen-log balance beams set over natural turf.

Sand Play Pits Organic-shaped sand zones bounded by timber edging for safe, tactile exploration.

Timber Climbing Frames Low-impact wooden structures that blend seamlessly into the landscape.

4.3 Northern Boundary: Slender Tree Demarcation

The northern edge is lined with a uniform row of **slender, conical evergreen trees** (e.g. *Thuja* or *Cypress*). This row serves a dual purpose: it provides structural framing and a green privacy screen for the park perimeter, while its high canopy and narrow profile preserve clear sightlines so parents can supervise the entire play zone from any point inside the park.

4.4 Southern Boundary: Sensory & Fruit Plantation

The southern edge is a diverse, vibrant buffer designed to offer children a rich sensory and interactive experience.

- **Flowering and fruit trees:** a curated mix of seasonal flowering trees (bright pink and white blooms) together with low-hanging fruit trees such as guava, mango and peach. The result is a shifting palette of colour, fragrance and an educational farm-to-table experience for visiting children.
- **Low hedges:** beneath the canopy, neatly manicured low shrubs and spherical hedges define the southern baseline, adding layers, texture and soft natural containment.

4.5 Circulation & Amenities

Pedestrian Walkway A linear, paved pathway runs parallel to the southern plantation zone, connecting the main entrance gate to the entire length of the park for easy stroller and wheelchair accessibility.

Rustic Seating Heavy-timber log benches are placed along the pathway and under the shade of the flowering trees, offering comfortable resting spots for parents and guardians with direct sightlines onto the central play areas.

4.6 Geometry & Layout: Curvilinear Play Areas

The internal zones, sand pits and pathways are deliberately designed with **organic, flowing, curvy geometries** rather than rigid straight lines.

- **Encouraging natural movement:** curving paths and edges mimic nature and encourage children to run, explore and play at a relaxed, intuitive pace.
- **Safety and softness:** eliminating sharp corners reduces collision hazards and makes the layout inherently safer for high-energy play.
- **Visual harmony:** the fluid shapes soften the long, narrow constraints of the site, making the park feel wider, more expansive and visually connected to the “Natural Playground” theme.

4.7 Recommended Species (Site 1 Edges)

Edge	Role	Indicative Species	Notes
Northern	Slender screen / visibility frame	<i>Thuja occidentalis</i> , <i>Cupressus sempervirens</i>	Uniform conical form, evergreen, 2–3 m spacing.
Southern (canopy)	Sensory & fruit	Guava (<i>Amba</i>), Peach (<i>Aruwa</i>), Mango (<i>Aanp</i>), seasonal flowering ornamentals	Low-hanging fruit safe for children; pink/white blooms.
Southern (ground)	Low hedge	<i>Buxus</i> spp., <i>Duranta</i> dwarf, <i>Murraya</i>	Spherical/clipped, 0.4–0.8 m height.

Table 2. Edge planting palette for the Children’s Park (Site 1).

5 Site 2: Community Forest — Edge Plantation

5.1 Concept

Site 2 is a cleared zone within the adjacent community forest, north of Dakchhinkali Road. It is developed as an **edge plantation** to protect future site functions while adding ecological value. Planting fruit or flowering trees along the perimeter creates an engaging sensory experience, while small hedges provide clear, natural demarcation between the cleared zone and the surrounding forest.



It is recommended to include edge plantation to protect future site functions while adding ecological value. Planting fruit or flowering trees along the perimeter will create an engaging sensory experience, while small hedges will provide clear, natural demarcation.

SITE 2 RECOMMENDATIONS

Figure 3. Site 2 edge plantation recommendation, shown as a three-band layering against the cleared zone in the community forest. The legend (right of figure) identifies: **yellow** = *pre-existing plantation* (the existing forest canopy, retained as-is), **pink** = *fruit / flowering tree strip* (the new mid-row to create an engaging sensory perimeter), and **green** = *hedges and small shrubs for site demarcation* (the outermost line, providing a clear natural boundary between the cleared interior and the forest beyond).

5.2 Recommended Species (Site 2)

Common	Scientific Name	Local Name	Spacing	Notes
Lapsi (Hog Plum)	<i>Choerospondias axillaris</i>	Lapsi	5 m	Indigenous, culturally significant.
Pear	<i>Pyrus communis</i>	Naspati	5–6 m	Reliable fruiting in mid-hills.
Peach	<i>Prunus persica</i>	Aruwa	5 m	Fruits within 2–3 years.
Plum	<i>Prunus salicina</i>	Aarubu	5 m	Hardy, attractive flowering.
Lychee	<i>Litchi chinensis</i>	Litchi	6 m	Thrives in Kathmandu Valley.
Guava	<i>Psidium guajava</i>	Amba	5 m	Soil-binding roots, quick fruiting.

Table 3. Site 2 fruit / flowering edge species.

Existing Forest Buffer. The surrounding community forest is preserved **as-is**. No clearing is permitted beyond the agreed Site 2 boundary. The dense surrounding canopy acts as a windbreak, maintains soil stability on the gorge slopes and provides natural ecological connectivity.

6 Site 3: Roadside Strip — Dakchhinkali Road

6.1 Constraints & Strategy

Site 3 is the narrow strip directly adjacent to Dakchhinkali Road. The road is planned for a future expansion to a **23 m width**, which will typically accommodate higher-speed and higher-volume traffic. Plantation immediately at the road edge may therefore be lost during expansion and could compromise driver visibility in the interim.



PLANTATION RECOMMENDATION FOR SITE 3

Figure 4. Site 3 plantation recommendation along Dakchhinkali Road. The annotated marker on the figure indicates the planned **23 m road (future expansion)** corridor, which will typically carry higher-speed and higher-volume traffic. The legend (right of figure) identifies **yellow** = *pre-existing plantation* (retained roadside trees) and **green** = *recommended shrub plantation, or no plantation at all*, kept low enough not to alter driver visibility. Because the right-of-way will be widened, trees planted close to the current road edge would be lost during expansion; a discontinuous low shrub line, or deferral of planting entirely, is therefore preferred over taller stock.

6.2 Recommendation

- **Preferred:** a discontinuous **low shrub** plantation that does not alter driver visibility and can be retained or relocated easily during road widening.
- **Alternative:** **no plantation at all** within the future right-of-way, deferring greening of this strip until after the road expansion is finalised.
- Pre-existing roadside trees (e.g. Jacaranda) remain undisturbed and are not affected by this recommendation.

7 Engineering Considerations

7.1 Root-System Management

All trees planted within **3 m** of the flagstone paving or shelter foundation must have **deep tap-root** systems (e.g. Guava, Lapsi, Sissoo) to prevent surface heaving. Shallow-rooted species such as *Eucalyptus* are explicitly excluded from this proposal.

7.2 Utility-Line Clearance

Utility and power lines run along the Dakchhinkali Road corridor directly adjacent to Site 3. Therefore:

- No shrub or tree in Site 3 shall exceed **1.5 m** mature height, and the future 23 m right-of-way must remain clear of any new planting.
- For Sites 1 and 2, trees must be planted at a minimum horizontal offset of **3 m** from any overhead line and shall not exceed 8 m mature height where lines are present.
- Tall species (Walnut, Alder, Sissoo), where used at all, must be set back well within the community forest, away from the road and utility corridor.

7.3 Slope & Drainage

The central slope has a natural gradient of approximately 5–8%. No grading or terracing is required. Trees planted on the downslope side should be staked securely for the first two growing seasons.

7.4 Right-of-Way & Road Expansion

The planned widening of Dakchhinkali Road to **23 m** is the principal external constraint on the design. Sites 1 and 2 sit *outside* the future right-of-way and are therefore unaffected; their mature trees will not require removal. Site 3 sits *within* the future right-of-way and is intentionally restricted to a low shrub line (or no planting) so that nothing of lasting value is lost when the road is widened. Any pre-existing roadside trees outside the new corridor are retained as-is.

7.5 Comparative Engineering Specification

Parameter	Site 1 (Children's Park)	Site 2 (Community Forest)	Site 3 (Roadside)
Max tree height	6–8 m (edges); play zone unplanted	8–10 m (no overhead lines)	≤1.5 m shrub line, or none
Root system	Deep tap-root preferred near paths	Any; broader roots acceptable	Shallow, removable shrubs only
Spacing	2–3 m (N edge); 4–5 m (S edge fruit)	5–6 m (orchard layout)	N/A (discontinuous shrubs)
Planting pit size	60 × 60 × 60 cm	60 × 60 × 60 cm	40 × 40 × 40 cm
Soil amendment	50% original + 30% compost + 20% coarse sand	50% original + 40% compost + 10% coarse sand	Minimal
Staking required	Yes, bamboo, 2 seasons	Yes, bamboo, 1–2 seasons	Not required
Road clearance	N/A (set inside park)	N/A	Min 1.5 m from future 23 m edge
Utility clearance	Min 3 m horizontal	Min 3 m horizontal	Min 1.5 m horizontal

Table 4. Engineering specification by site.

8 Technical Methodology for Fruit Plantation

Fruit trees require greater initial care, precise spacing, and nutrient-rich soil than standard forest trees in order to ensure survival and future fruiting.

8.1 Phase 1: Site Layout & Spacing

A square or staggered triangular grid is recommended for the Site 2 orchard. The same spacing logic applies to the fruiting species on the southern edge of Site 1. Crowding fruit trees reduces sunlight and yields and increases pest vulnerability.

- Guava, Peach, Plum, Citrus: 4 m × 4 m (~13 ft × 13 ft).
- Lychee: at least 6 m × 6 m due to its large mature canopy.

8.2 Phase 2: Pit Excavation & Soil Conditioning

1. **Pit size:** dig larger pits than for forest stock, 0.75 m × 0.75 m × 0.75 m.
2. **Layering & conditioning:** mix excavated topsoil with organic compost / farmyard manure (FYM) and *neem cake* (*Neem ko Khali*) in a 1:1 ratio. Neem cake is critical to prevent termites and root-rot diseases.
3. Add a handful of bone meal or phosphate to encourage rapid initial root development.

8.3 Phase 3: Plantation Execution (Environment Day)

1. **Graft union management:** most quality fruit saplings (citrus, peach, plum) are grafted. Ensure the graft union (the bump on the lower stem) sits at least **2–3 inches above** the final soil level. Burying

the graft union causes the tree to rot or lose its fruiting characteristics.

2. **Basin preparation:** create a wide, ring-like soil depression (*thāpalā*) of radius ~ 0.5 m around each tree to act as a water-harvesting basin.
3. **Mulching:** immediately after watering, cover the basin with a 2-inch layer of dry grass or straw. Mulching retains soil moisture, suppresses weeds, and regulates soil temperature.

8.4 Phase 4: Long-Term Sustainability & Protection

- **Tree guards:** sturdy, tall bamboo tree guards are mandatory. Without them, saplings will almost certainly be trampled or damaged within weeks.
- **Handover protocol:** as a Worldlink CSR event, a **1-year maintenance protocol** should be formalised with the Manjushree Park caretakers and CFUG. Fruit trees require regular watering through the first winter dry season to survive until the next monsoon.

9 Park Design Recommendations (Supporting Notes)

The following items complement the Site 1 Children's Park design (Section 4) and the Site 2 edge plantation (Section 5). Where any item below conflicts with the Children's Park design, the Children's Park design prevails.

9.1 Land Preparation

Level the uneven grassy field to create a clean base for pathways, seating, and plantations; retain some natural contours where possible for character and drainage.

9.2 Green Wall & Boundary Treatment

Use climbing plants (e.g. ivy, bougainvillea, local creepers) along walls and fences to soften hard edges and blend the park with the surrounding landscape.

9.3 Plantation Strategy

- Avoid very large canopy trees that could block views or reduce usable space.
- Opt for medium-sized ornamental trees (Jacaranda, Gulmohar, Amaltas) and flowering shrubs for seasonal colour.
- Introduce butterfly-attracting plants (lantana, milkweed, marigold) in clusters.
- Where space allows, plant additional fruit-bearing trees (guava, pomegranate, citrus) for ecological and community value.

9.4 Pathways & Circulation

Lay permeable stone pathways to reduce water runoff and maintain a natural look. Curved paths can guide visitors toward seating areas, gazebos, or scenic viewpoints.

9.5 Seating & Relaxation

Construct shaded sitting areas (gazebos, pergolas, benches under small trees) strategically placed near plantations for a cool, relaxing environment. A central lawn can support yoga, picnics, or small gatherings.

9.6 Wildlife-Friendly Features

Install birdhouses on trees or poles to attract local species, and create small ecological pockets with flowering plants to encourage butterflies and pollinators.

9.7 Water & Sustainability

- Where feasible, add a small pond or fountain to enhance tranquillity and support aquatic plants.
- Integrate rainwater harvesting for irrigation.
- Provide composting bins to recycle organic waste into fertiliser for the plantations.

9.8 Information & Awareness

Place plant information boards to educate visitors about species and their benefits. Display park rules at entry points to ensure cleanliness, safety, and respect for nature.

10 Summary & Approvals

CFUG Approval. All three sites (Site 1 Children's Park, Site 2 Community Forest edge, Site 3 Roadside strip) have been confirmed by the CFUG President for the Worldlink Environment Day 2026 plantation.

Deliverables. On Environment Day, the implementation team will:

1. Establish Site 1 Children's Park: install log balancing trails, sand pits and timber climbing frames on curvilinear turf islands; plant the slender-tree northern boundary, the flowering / fruit southern boundary and the low spherical hedge line; lay the paved walkway and place rustic log benches.
2. Plant Site 2 with the fruit / flowering edge species (Lapsi, Pear, Peach, Plum, Lychee, Guava) and a hedge line for demarcation.
3. Limit Site 3 to a low shrub line (or defer planting entirely) within the future 23 m road right-of-way.
4. Install bamboo guards and water basins around each sapling.
5. Hand over a 1-year maintenance protocol to the CFUG and park caretakers.

Expected outcome. A safe, attractive, low-maintenance three-site plantation centred on the Natural Playground at Site 1, supported by the Site 2 sensory edge plantation and a sightline-respecting Site 3 roadside treatment, that meets road-safety and utility-clearance constraints, supports local biodiversity, and provides an educational, edible landscape for the Chobhar community.